DEEP LEARNING IN NETWORK TRAFFIC ANALYSIS USING SYNTHETIC DATA FOR PRIVACY PROTECTION AND CYBER ATTACK MITIGATION

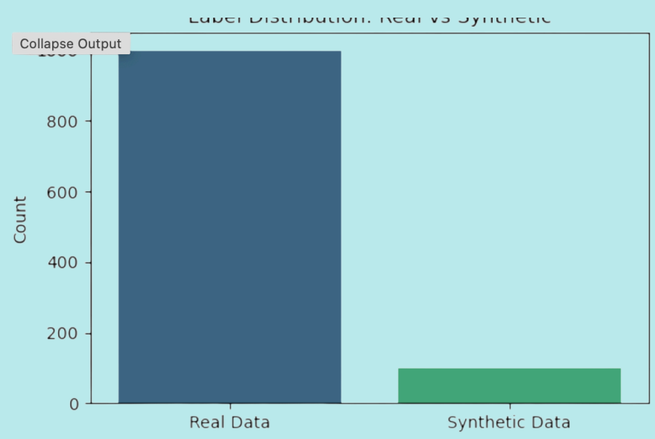
Swikriti Neupane , Dr Pratap Sahu

UR2PhD

Undergraduate Research to PhD Mentoring

Background

- Intrusion Detection Systems (IDS):
  Used to detect abnormal or malicious network activity. Depend on large, labeled datasets for training.



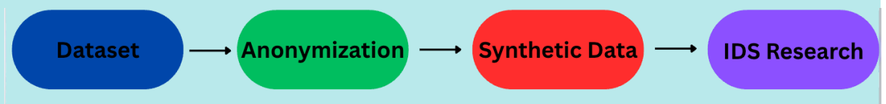
- **Challenge with Real Data:**
  Contains Personally Identifiable Information (PII) such as IP addresses, hostnames, and user credentials.Sharing such datasets directly creates privacy and security risks.
- **PII Anonymization:**
  Applied tokenization (e.g., IP → IP_001) and generalization (e.g., exact timestamp → date). Ensures sensitive details are hidden before dataset use.
- **Synthetic Data Generation:**
  Implemented CTGAN and related models to create synthetic traffic that mimics real distributions.
  Evaluated on three key factors:
  - Utility – Does synthetic data support model training without accuracy loss?
  - Diversity – Are generated samples varied enough to prevent overfitting?
  - Privacy – Are synthetic records sufficiently different from real ones?

Objective

- Remove personally identifiable information (PII) from network datasets while maintaining their utility.
- Generate high-quality synthetic datasets using GAN-based models.
- Validate synthetic data on utility, diversity, and privacy metrics.
- Enable secure data sharing for IDS research without compromising sensitive information.

Methodology

01 Dataset: Used the CIC-IDS-2017 dataset because it reflects both real network activity and different types of cyberattacks.



02 PII Detection: Applied pattern-matching techniques to locate personal data such as IPs, MAC addresses, hostnames, and URLs.

03 Anonymization:
- Tokenization – replaced sensitive values with coded labels.
- Generalization – made details broader (e.g., specific IPs → network ranges; timestamps → dates)

04 Synthetic Data Generation Models:
- CTGAN and TVAE for network flow data.
- GPT-2 for text-based logs.
- Diffusion models for image-based experiments.

05 Evaluation:
- Utility – Compared patterns of real and synthetic data using KDE, PCA, and t-SNE.
- Diversity – Checked that synthetic data had enough variety.
- Privacy – Measured how hard it is to match synthetic data back to the original data.

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Future Direction

- Advanced Models: Explore GNNs, diffusion for tabular, LLMs for logs.
- Automated Pipelines: End-to-end anonymization + synthetic data generation.
- Fairness: Balance rare vs. frequent attack classes

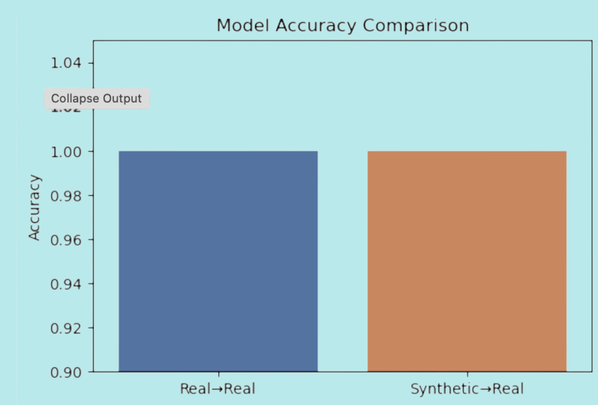
Conclusion

Synthetic data generation can protect privacy while preserving research value. Our pipeline achieved:
- 100% accuracy with no performance drop.
- Consistent label distribution across datasets.
- Strong privacy protection (large NN distances).
- This shows synthetic data can support IDS research without exposing sensitive information.

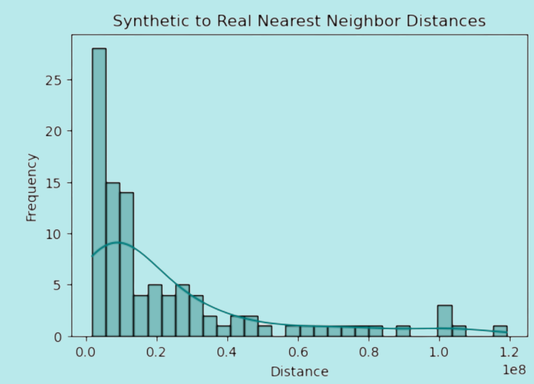
Results

To evaluate the effectiveness of the synthetic data pipeline, we compared real vs. synthetic datasets across three dimensions: Utility, Diversity, and Privacy.

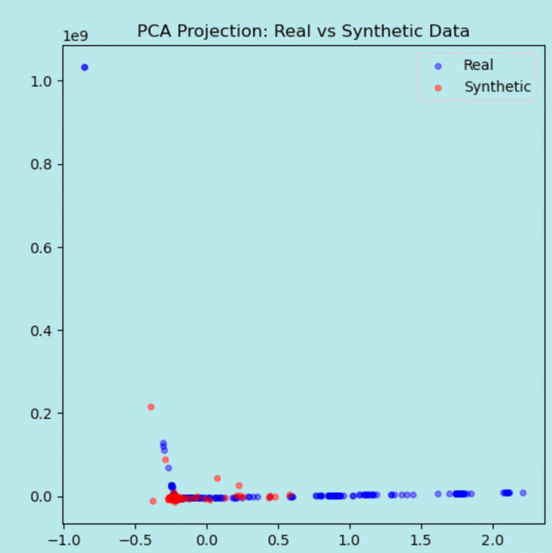
- Utility (Preserving Model Performance):
  ML model trained on both real and synthetic data achieved same accuracy i.e 100% showing no accuracy drop (Δ =0.0000)



- Diversity (Variation in Synthetic Data):
  Label distribution between real and synthetic datasets was consistent. Synthetic data generated sufficient unique samples rather than repeating originals.



- Privacy (Protecting Sensitive Records):
  Average distance (synthetic → nearest real sample): 23,952,242.9681 Minimum distance (closest synthetic → real): 2,583,688.0567 which indicates synthetic data is not direct copy of real data since the distance is large.



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